

Explainable Multimodal Deep Learning for Predicting Thyroid Cancer Lateral Lymph Node Metastasis Using Ultrasound Imaging

Corresponding Author: Dr Siqiong Yao

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

This paper introduces LLNM-Net, a deep learning model designed to address the challenges of detecting lateral lymph node metastasis (LLNM) in thyroid cancer. LLNM-Net leverages a bidirectional attention architecture to integrate multimodal data, including imaging, clinical reports, and demographic details from a cohort of 23,692 patients and evaluated on a test cohort of 9,836 patients across five centers. Demonstrating superior predictive accuracy (AUC 0.944, accuracy 0.847) for metastasis compared to human experts, it identifies high-risk patients with a precision AUC of 0.983. Utilizing interpretable methods, the model highlights critical risk factors and susceptible regions, offering a transformative tool for preoperative LLNM screening and personalized treatment planning in thyroid cancer care.

This paper establishes a well-substantiated motivation for addressing a critical clinical challenge: predicting metastasis in thyroid cancer. The ability to accurately predict lateral lymph node metastasis (LLNM) holds significant clinical value, as it directly impacts treatment planning, prognosis, and recurrence risk management.

Comments:

1. The study lacks a thorough comparison with existing baseline models commonly used for metastasis prediction. Conducting extensive benchmarking against these established methods would provide greater context for evaluating the performance and significance of the proposed approach.
2. The authors should also provide comprehensive ablation studies to demonstrate the necessity and effectiveness of each component in the proposed model (Fig. 1). For instance, why bidirectional attention? Why the Central Point Distance Transformation (CPDT) method?
3. The proposed model heavily relies on the U-Net++ and YOLOv8 architectures for obtaining segmentation masks and object locations. However, the authors have not reported the performance metrics of these models.
4. In practical clinical scenarios, are these segmentation and detection models truly robust enough? Segmentation and object detection in medical imaging are inherently challenging tasks due to the high variability and complexity of clinical data, making it crucial to evaluate their reliability and consistency.
5. What is the specific contribution of each data modality to the final decision-making process in the model? It would be insightful to see how the model performs when each type of data is used independently, i.e., imaging, clinical reports, and demographic details.
6. The comparison between LLNM-Net and human experts seems unfair due to the differing dataset sizes. Could you provide the predictive performance of LLNM-Net on the smaller dataset (N=200) used to evaluate the radiologists? This would allow for a more equitable and accurate comparison.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

The authors present a machine learning model that predicts the presence of lateral neck lymph node metastases from thyroid cancer. The performance of the classifier is impressive, and the quality of data science is high. Using AI to identify locations within the thyroid more likely to be associated with metastatic disease is an interesting and useful result.

I have questions about the clinical utility and intended use of the classifier. In my opinion, it should not be used to guide a decision to perform lateral neck dissection as it will result in overuse of this procedure and likely harm to patients with low-volume lateral neck metastatic disease.

Major concerns:

1. Detecting CLINICAL (detected on physical examination or radiology assessment) LLNMs is important, and neck dissection is indicated by the current clinical guidelines. However, predicting microscopic LLNMs with AI may be harmful because the removal of these small metastases may not change the treatment outcome. In this scenario, using the AI model will result in overutilization of lateral neck dissection and patient harm. Please provide features of the LLNMs (size, number, and presence of extranodal extension) for all groups (including those missed by radiologists). The poor performance of radiologists could be due to the small size of "missed" LLNMs, which may not be clinically significant.
2. 18,767 patients did not have metastatic disease (please clarify if Table 1 refers to lateral, central or both lateral and central neck LN metastases; if both please quantify those separately) but had lateral neck dissection. Why were these patients operated on? These are unnecessary neck dissections.
3. Decision curve analysis in 6 c,d is based on the assumption that all LLNDs based on true AI predictions are indicated and beneficial to the patient. As outlined in #1, this is probably not true for patients with microscopic metastases. Due to these considerations, I do not think that the workflow in 7b would be the best clinical management. It is OK to use AI output to have a lower threshold for FNA of the indeterminate lateral neck LNs.
4. Lines 343-348. LLNM-Net is trained on cancerous nodules only. Most of the thyroid nodules in the population will be benign. Therefore, LLNM-Net cannot be utilized for population screening. Also, detecting lymph node metastases >3 cm in size is trivial and radiologists would likely achieve close to 100 % sensitivity on such a task. Finally, what is the negative class in Table 2: Both no metastatic disease and < 3 cm metastatic disease?
5. Fig 7a. The best clinical practice is to evaluate central and lateral neck lymph nodes before the FNA (and biopsy suspicious LNs at the time of the procedure, see Recommendation 6 in the ATA guidelines). Also, lateral neck evaluation may change decisions on whether to proceed with thyroidectomy (i.e., the presence of metastatic LNs in the lateral neck would require total thyroidectomy).

Minor considerations.

1. Fig 4c. Please label left and right. Is it a radiologist's view (the right is on the left) or a conventional view?

(Remarks on code availability)

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Thank the authors for addressing the issues raised during the first round of review. The revisions have improved the clarity and quality of the manuscript.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

The authors did a significant amount of additional work and analyses to address concerns raised at the initial review.

The clinical workflow shown in Fig. 7 has improved, and the AI model is now used to guide the decisions to evaluate cervical LNs with ultrasound and to proceed with the FNA. I argue that ALL patients evaluated for thyroid nodules should get an ultrasound assessment of central and lateral neck lymph nodes before the decision to proceed with FNA is made. However, I understand that the differences could be due to practice patterns specific to China, and I do not feel that the publication of this otherwise very interesting study should be delayed further.

(Remarks on code availability)

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REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author): Expert in machine learning and deep learning for medical image analysis

This paper introduces LLNM-Net, a deep learning model designed to address the challenges of detecting lateral lymph node metastasis (LLNM) in thyroid cancer. LLNM-Net leverages a bidirectional attention architecture to integrate multimodal data, including imaging, clinical reports, and demographic details from a cohort of 23,692 patients and evaluated on a test cohort of 9,836 patients across five centers. Demonstrating superior predictive accuracy (AUC 0.944, accuracy 0.847) for metastasis compared to human experts, it identifies high-risk patients with a precision AUC of 0.983. Utilizing interpretable methods, the model highlights critical risk factors and susceptible regions, offering a transformative tool for preoperative LLNM screening and personalized treatment planning in thyroid cancer care.

This paper establishes a well-substantiated motivation for addressing a critical clinical challenge: predicting metastasis in thyroid cancer. The ability to accurately predict lateral lymph node metastasis (LLNM) holds significant clinical value, as it directly impacts treatment planning, prognosis, and recurrence risk management.

Comments:

1. The study lacks a thorough comparison with existing baseline models commonly used for metastasis prediction. Conducting extensive benchmarking against these established methods would provide greater context for evaluating the performance and significance of the proposed approach.

Response:

Thank you for your suggestion. We did perform a comparison with baseline models, and we have subsequently supplemented the comparative results with the latest models. The detailed results can be found in Table S7 of the supplementary materials.

Line 64 in supplementary appendix:

“

Table S7 Comparison of prediction performance among different AI methods on validation sets.

Method ^a	AUC	Spe.	Sen.	Acc.	NPV	PPV
Early fusion ¹	0.776	0.653	0.767	0.713	0.720	0.707
Late fusion ²	0.799	0.702	0.772	0.737	0.750	0.736
GPT-4V ³	0.733	0.688	0.713	0.701	0.719	0.694
Qwen2.5-VL ⁴	0.685	0.634	0.647	0.641	0.655	0.641
Deepseek-VL2 ⁵	0.722	0.649	0.702	0.676	0.707	0.669
Gemini-Pro ⁶	0.712	0.672	0.621	0.647	0.645	0.670

IRENE ⁷	0.874	0.724	0.851	0.790	0.818	0.769
MMC-Net ⁸	0.832	0.756	0.761	0.758	0.820	0.684
DeepThy-Net ⁹	0.785	0.714	0.734	0.723	0.779	0.662
Inception V3 ¹⁰	0.777	0.624	0.792	0.716	0.709	0.722
ACE-Net ¹¹	0.811	0.664	0.807	0.741	0.748	0.736
Image-only	0.863	0.850	0.702	0.771	0.714	0.843
LLNM-Net	0.948	0.946	0.841	0.890	0.839	0.947

a. All methods utilized the same data preprocessing, model tuning techniques, etc.

”

2. The authors should also provide comprehensive ablation studies to demonstrate the necessity and effectiveness of each component in the proposed model (Fig. 1). For instance, why bidirectional attention? Why the Central Point Distance Transformation (CPDT) method?

Response:

We appreciate your identification of this point. By incorporating the bidirectional attention module, we explicitly encode the interconnections between different modalities, bridging the gap from modality-specific features at the symbolic level to holistic representations at higher levels of the model aimed at prediction. This complements the multimodal learning process of relevant features across modalities, enabling better handling of inter-modality discrepancies. Through the Central Point Distance Transformation (CPDT) method, we convert distance information, which the model cannot directly interpret, into recognizable grayscale values, standardizing it to facilitate the extraction of the relative positional relationship between nodules and thyroid edges. We have conducted ablation experiments, and the results for multiple components, including the bidirectional attention module, CPDT method, and experiments involving different data, are presented in Table S8 of the supplementary materials.

Line 74 in supplementary appendix:

“

Table S8 An ablation study of LLNM-Net on validation sets, removing or replacing individual components.

Index	Location	CPDT	Morphology	Radiological reports	Demographics	AUC
BMA(0)	✓	✓	✓	✓	✓	0.799
BMA(2)	✓	✓	✓	✓	✓	0.912
BMA(4)	✓	✓	✓	✓	✓	0.948
BMA(6)	✓	✓	✓	✓	✓	0.927
Unidirection	✓	✓	✓	✓	✓	0.843
BMA(4)			✓	✓	✓	0.729
BMA(4)	✓		✓	✓	✓	0.793
BMA(4)	✓	✓		✓	✓	0.814
BMA(4)	✓	✓	✓		✓	0.860
BMA(4)	✓	✓	✓	✓		0.939

BMA(N) indicates that there are a total of 16 blocks in LLNM-Net, among which N are bidirectional attention blocks, and the remaining blocks are self-attention blocks. Location and Morphology are features extracted from the images. CPDT means that the location feature map extracted from the image using the CPDT method is used as input, otherwise the image itself is used as input. Unidirection means that we only compute attention from text to image in the bidirectional attention blocks. Radiological reports represent the corresponding report texts for the images, and Demographics represent patient-related clinical information. For each row in the table, check marks indicate that the modules are used in the model, and the absence of a check mark indicates that the related module has been removed. The evaluation metric used is AUC.

”

3. The proposed model heavily relies on the U-Net++ and YOLOv8 architectures for obtaining segmentation masks and object locations. However, the authors have not reported the performance metrics of these models.

Response:
Thank you for your suggestion. We have included the segmentation performance metrics for these models in the supplementary materials. Please refer to Table S4 and Table S5 for the detailed results.

Line 51 in supplementary appendix:
“

Table S4 The performance of segmentation module.

Datasets ^a	Training set N=5816	Validation set N=1454	Test set N=6657
Thyroid Segmentation (mIoU)	0.845	0.827	0.818
Nodule Segmentation (mIoU)	0.883	0.848	0.841

a. The table displays performance of segmentation at the image level, where "N" in the table represents the number of images annotated by the radiologists.

Table S5 The performance of segmentation module in test set.

Thyroid cancer subtypes ^a	PTC N=4143	FTC N=1806	MTC N=708
Thyroid Segmentation (mIoU)	0.818	0.820	0.815
Nodule Segmentation (mIoU)	0.835	0.856	0.840

a. The table displays performance of segmentation at the image level, where "N" in the table represents the number of images annotated by the radiologists. The test set contains imaging data from four medical centers.

”

4. In practical clinical scenarios, are these segmentation and detection models truly robust enough? Segmentation and object detection in medical imaging are inherently challenging tasks due to the high variability and complexity of clinical data, making it crucial to evaluate their

reliability and consistency.

Response:

We appreciate your constructive feedback. To validate the generalization performance of our segmentation model, we additionally collected a dataset of 6,657 thyroid cancer images from 2,863 patients across four external independent centers for validation. These datasets included one conventional subtype and two rare subtypes of thyroid cancer. The results for thyroid margin segmentation and internal tumor segmentation show no significant difference in performance compared to the primary center's results ($P < 0.01$). Detailed results are provided in Table S4 and Table S5 of the supplementary materials.

Line 51 in supplementary appendix:

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a. The table displays performance of segmentation at the image level, where "N" in the table represents the number of images annotated by the radiologists. The test set contains imaging data from four medical centers.

”

5. What is the specific contribution of each data modality to the final decision-making process in the model? It would be insightful to see how the model performs when each type of data is used independently, i.e., imaging, clinical reports, and demographic details.

Response:

Thank you for your thoughtful advice. In the main text, we have illustrated the contributory role of each feature toward predicting metastasis outcomes, such as positional information within imaging, feature information, and clinical data (Fig. 5). Furthermore, we have supplemented the ablation results for each type of data in relation to prediction in Table S8, and in Table S9, we have presented the predictive performance of each data modality when used independently for forecasting.

Line 286 in manuscript:

“

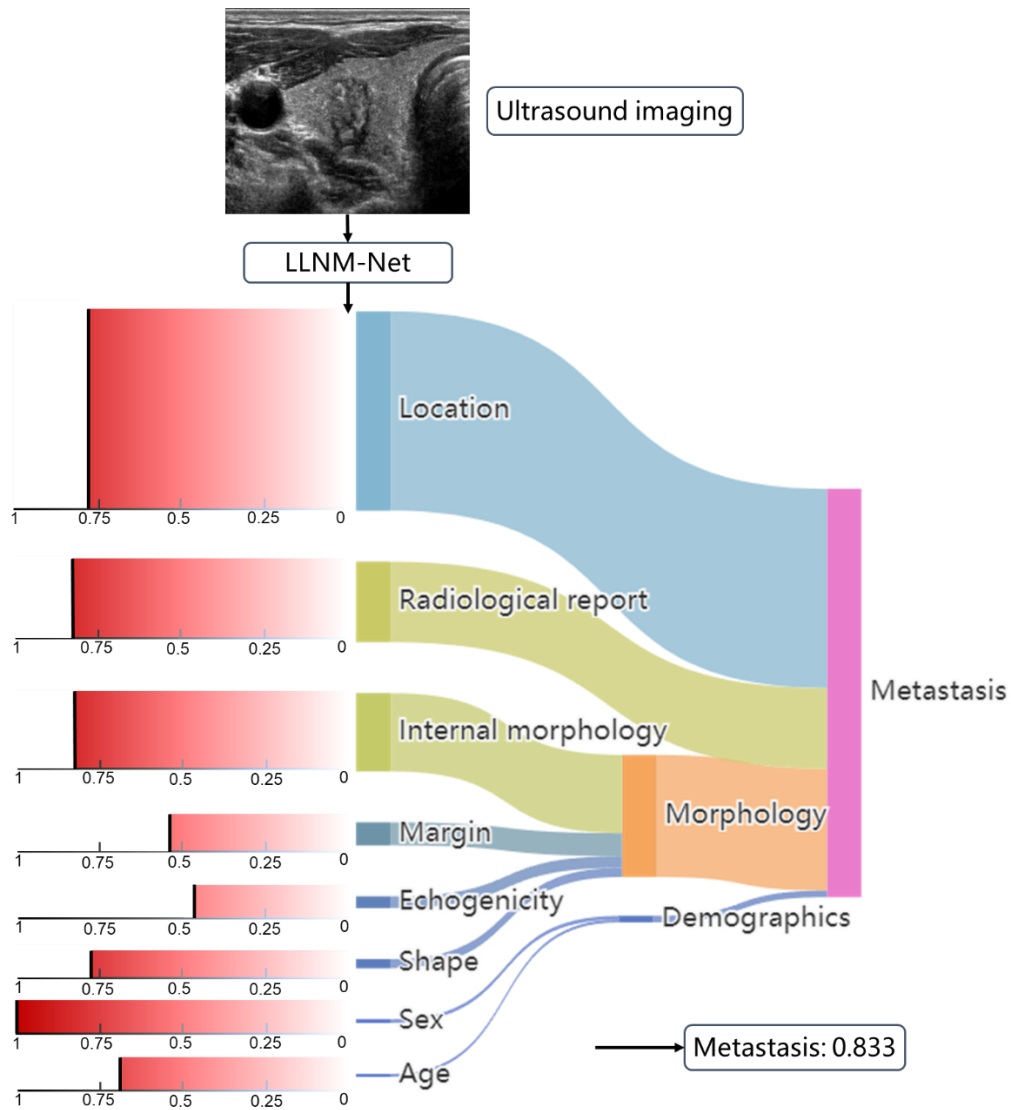


Fig 5. An illustrative example of LLNM-Net results.

”

Line 74 in supplementary appendix:

“

Table S8 An ablation study of LLNM-Net on validation sets, removing or replacing individual components.

Index	Location	CPDT	Morphology	Radiological reports	Demographics	AUC
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BMA(2)	✓	✓	✓	✓	✓	0.912
BMA(4)	✓	✓	✓	✓	✓	0.948
BMA(6)	✓	✓	✓	✓	✓	0.927
Unidirection	✓	✓	✓	✓	✓	0.843
BMA(4)			✓	✓	✓	0.729
BMA(4)	✓		✓	✓	✓	0.793
BMA(4)	✓	✓		✓	✓	0.824

BMA(4)	✓	✓	✓	✓	0.860
BMA(4)	✓	✓	✓	✓	0.939

”

Line 87 in supplementary appendix:

“

Table S9 The results when using a certain modality data independently..

Index	Image		Radiological reports	Demographics	AUC
	Location	Morphology			
CPDT+CNN	✓				0.797
CNN		✓			0.692
CPDT+CNN	✓	✓			0.825
Transformer			✓		0.646
GLM				✓	0.617

The results of using a specific data type for independent prediction, where different data types are classified using appropriate methods. For example, using CNN for image data, transformer models for text data, and generalized linear equations (GLM) for demographics data.

”

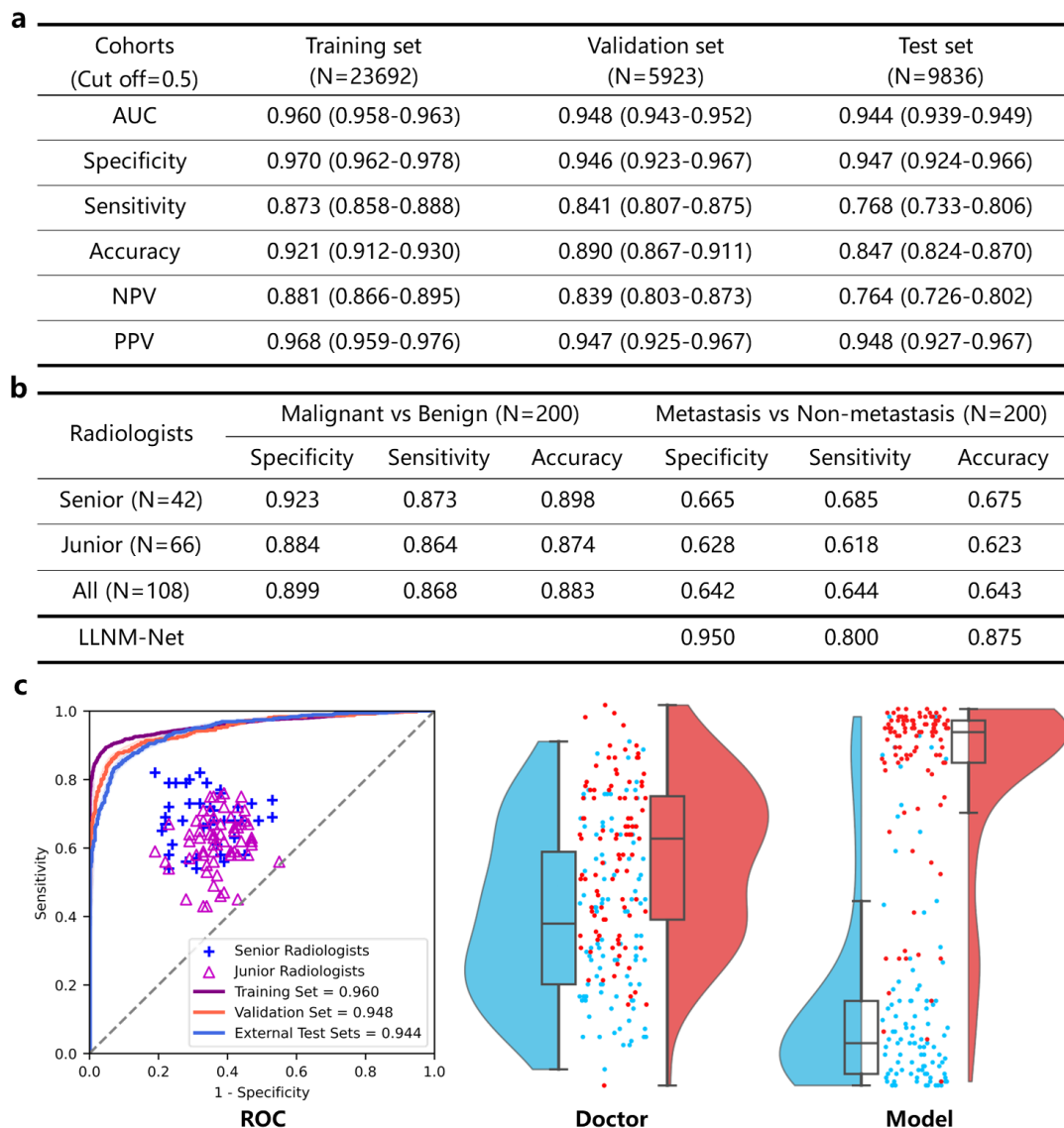
6. The comparison between LLNM-Net and human experts seems unfair due to the differing dataset sizes. Could you provide the predictive performance of LLNM-Net on the smaller dataset (N=200) used to evaluate the radiologists? This would allow for a more equitable and accurate comparison.

Response:

Thank you for pointing that out. We have provided the predictive performance of LLNM-Net on a small dataset (N=200), and the results can be found in Fig. 3b:

Line 234 in Manuscript:

“



”

Line 239 in Manuscript:

“

b. Prediction performance of the comparative test between radiologists and LLNM-Net. Radiologists performed well in the malignant classification test but poorly in the LLNM classification test. In contrast, LLNM-Net performed better on the same test dataset.

”

Reviewer #2 (Remarks to the Author): Clinical expert for thyroid cancer ultrasound imaging and metastases, and AI

The authors present a machine learning model that predicts the presence of lateral neck lymph node metastases from thyroid cancer. The performance of the classifier is impressive, and the quality of data science is high. Using AI to identify locations within the thyroid more likely to be associated with metastatic disease is an interesting and useful result.

I have questions about the clinical utility and intended use of the classifier. In my opinion, it

should not be used to guide a decision to perform lateral neck dissection as it will result in overuse of this procedure and likely harm to patients with low-volume lateral neck metastatic disease.

Major concerns:

1. Detecting CLINICAL (detected on physical examination or radiology assessment) LLNMs is important, and neck dissection is indicated by the current clinical guidelines. However, predicting microscopic LLNMs with AI may be harmful because the removal of these small metastases may not change the treatment outcome. In this scenario, using the AI model will result in overutilization of lateral neck dissection and patient harm. Please provide features of the LLNMs (size, number, and presence of extranodal extension) for all groups (including those missed by radiologists). The poor performance of radiologists could be due to the small size of “missed” LLNMs, which may not be clinically significant.

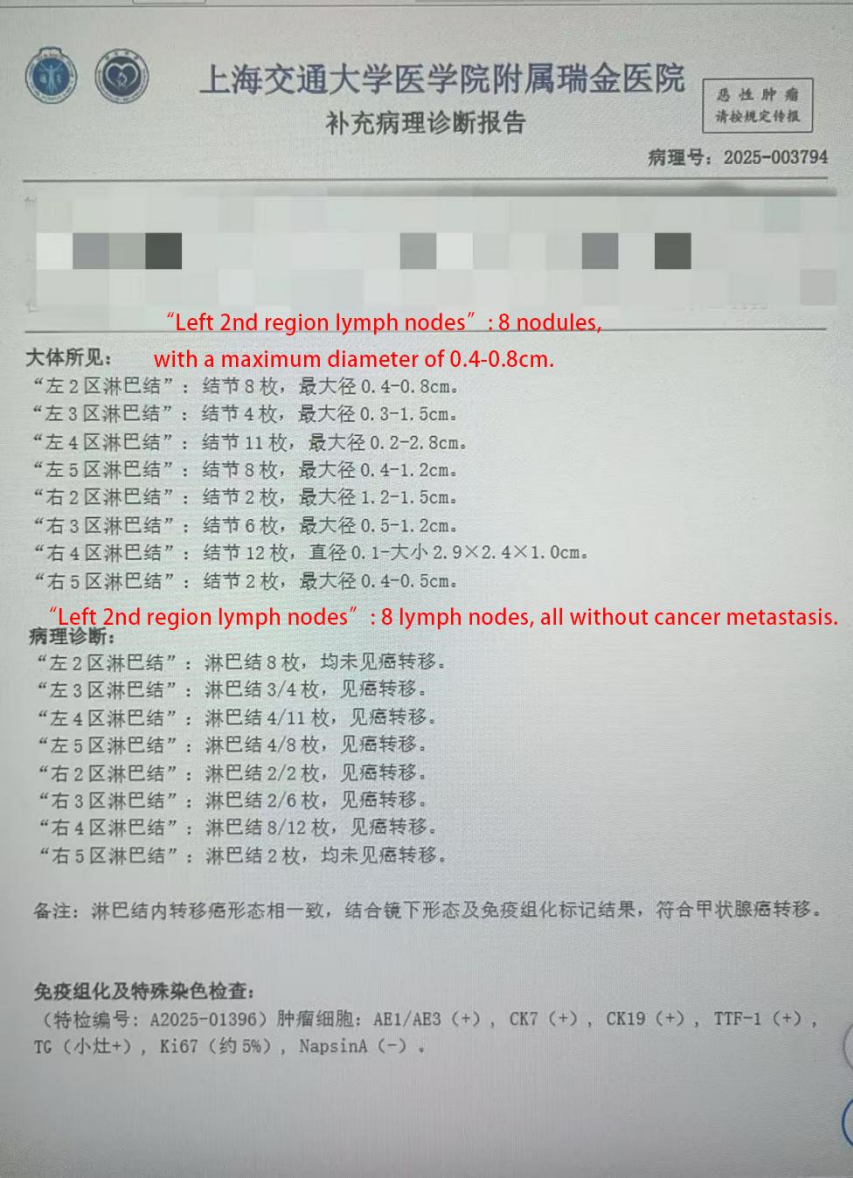
Response:

Your clinical insights have significantly deepened our understanding of the practical application of AI in clinical diagnosis—thank you very much for this! In response, we have supplemented the LLNM characteristics for all groups (Table S3) and would like to further discuss our approach. The original intent behind the design of our AI analysis and prediction system for lateral lymph node metastasis was to provide preoperative assistance to clinical experts, thereby enhancing their decision-making capabilities. Our system estimates the probability of metastatic risk and offers a decision-making reference for low, medium, and high metastasis risk. Microscopic LLNMs are classified as low-risk factors, suggesting that doctors conduct neck ultrasound examinations and make their own judgment on whether to proceed with further examinations, such as puncture or other treatments; it is not intended to directly lead to overutilization.

For example, the model highlights key imaging features or clinical high-risk factors that contribute to prediction. We hope that clinical experts can make more informed and reasonable decisions by combining their own experience with the AI system's judgment bases, which compensates for cognitive limitations and prevents blind reliance on AI.

In fact, our training data were provided by clinical experts with over 5 years of experience in top-tier hospitals in China. Positive pathology cases were confirmed by ultrasound, puncture pathology, and lymph node pathology, while negative cases were those with negative puncture pathology (with no excessive lateral lymph node dissection). However, since the previous comparative experiment only utilized 100 transfer data, we conducted an additional experiment in this revision to investigate missed diagnoses by doctors with varying levels of experience, using a larger dataset. We invited 20 doctors with 1-3 years of clinical experience from top-tier hospitals, as well as 20 doctors from lower-tier hospitals and community clinics, to assess 2,000 patient data (including 1,000 transferred patients and 1,000 non-transferred patients) and determine if lateral lymph node metastasis had occurred. The results showed more than 30% missed diagnoses, and the average maximum diameter of missed positive lymph nodes was smaller than the overall average. However, some patients with larger positive lymph nodes, especially those located posteriorly, were still missed, as they were difficult to diagnose with ultrasound. Please refer to Table S3 for details.

The paraffin pathology reports included in this study did not routinely report the presence of extranodal extension or the size of individual positive lymph nodes, but only the maximum and minimum lymph node sizes, as well as the Mean Harvested LLN, Mean Positive LLN, and Mean Positive LN Maximum Diameter. Therefore, extranodal extension information is not included in Table S3. For clarity, we have attached a case from the main center’s pathology report (with two red English sentences added for ease of understanding in the figure).



Line 3 in supplementary appendix:
“

Radiologist competing test

We conducted an extra evaluation test to assess the number of lymph nodes missed by doctors. We involved 20 doctors with 1-3 years of experience from top-tier hospitals and 20 from lower-tier hospitals and community clinics. Using ultrasound images, imaging reports, and clinical

information from 2000 patients in our dataset, they predicted whether lateral lymph node metastasis would occur. The dataset comprised 1000 LLNM-positive and 1000 LLNM-negative cases. We analyzed the average lymph node data from the group of patients missed by the doctors, as detailed in Table S3.

”

Line 48 in supplementary appendix:

“

Table S3 LLNMs information of different groups in the comparative test.

	Patients	Mean Harvested LLN	Mean Positive LLN	Mean positive LN maximum diameter
Total	20684	24.5(11-56)	6.3(1-41)	1.08(0.2-3.5)
Test data	1000	23.3(11-37)	6.2(1-32)	0.97(0.2-3.4)
Radiologists recognition ^a	653	24.7(11-37)	6.3(1-32)	1.06(0.2-3.4)
Radiologists missing ^a	347	20.7(11-37)	6.0(1-32)	0.80(0.2-3.4)
AI recognition	955	23.4(11-37)	6.2(1-32)	0.97(0.2-3.4)
AI missing	45	21.2(12-31)	6.2(1-26)	0.95(0.2-3.1)

a. The table presents the mean values of identifications made by all doctors for 1000 patients.

”

2.18,767 patients did not have metastatic disease (please clarify if Table 1 refers to lateral, central or both lateral and central neck LN metastases; if both please quantify those separately) but had lateral neck dissection. Why were these patients operated on? These are unnecessary neck dissections.

Response:

Thank you for your feedback. We have clarified the regions of metastasis (central and lateral) in Table 1. All 18,767 patients were confirmed to have no lateral lymph node metastasis (LLNM) through puncture biopsy and did not undergo lateral neck dissection. We apologize for the incomplete description. In our country, routine central lymph node dissection is performed; therefore, "received surgery" refers to "central lymph node dissection." We have made the corresponding revisions in the manuscript.

Line 883 in Manuscript:

“

Table 1 Characteristics of the patients at baseline.^a

Characteristic	Training sets	Validation sets	External test sets	Total
Participants, n	23692	5923	9836	39451
Cancer participants, n (%)				
LLNM	12036 (51)	3158 (53)	5490 (56)	20684 (52)
Only LLNM	1769 (7)	558 (9)	826 (8)	3153 (8)
CLNM and LLNM	10267 (43)	2600 (44)	4664 (47)	17531 (44)
No LLNM	11656 (49)	2765 (47)	4346 (44)	18767 (48)

Only CLNM	6695 (28)	1512 (26)	1965 (20)	10172 (26)
No LNM	4961 (21)	1253 (21)	2381 (24)	8595 (22)
Age, years, median (IQR)	43 (34-54)	43 (35-53)	45 (36-51)	43 (35-52)
Sex, n (%)				
Female	16773 (71)	4079 (69)	7638 (78)	28490 (72)
Male	6919 (29)	1844 (31)	2198 (22)	10961 (28)
Nation, n (%)				
Han nationality	22491 (95)	5767 (97)	7569 (77)	35827 (91)
Minority nationality ^b	1201 (5)	156 (3)	2267 (23)	3624 (9)
Kwak TIRADS, n (%)				
1-3	116 (0)	34 (1)	141 (1)	291 (1)
4 (4A-4C)	21473 (91)	5360 (90)	7648 (78)	34481 (87)
4A	1285 (5)	294 (5)	3044 (31)	4623 (12)
4B	12871 (54)	3440 (58)	2661 (27)	18972 (48)
4C	7317 (31)	1626 (27)	1943 (20)	10886 (28)
5-6	2103 (9)	529 (9)	2047 (21)	4679 (12)
Metastatic number, n (%)				
0	11656 (49)	2765 (47)	4346 (44)	18767 (48)
1-5	10102 (43)	2535 (43)	4867 (49)	17504 (44)
6-10	1248 (5)	454 (7)	442 (4)	2144 (6)
>10	686 (3)	169 (3)	181 (2)	1036 (3)
Max tumor size, mm, n (%)				
<10	17404 (73)	4220 (71)	7079 (72)	28703 (73)
>10	6288 (37)	1703 (29)	2757 (28)	10748 (27)
Histological subtypes, n (%)				
Papillary Thyroid Cancer	21977 (93)	5171 (87)	8656 (88)	35804 (91)
Follicular Thyroid Cancer	1351 (6)	635 (11)	859 (9)	2845 (7)
Medullary Thyroid Cancer	364 (2)	117 (2)	321 (3)	802 (2)

a. Percentages may not add up to 100 due to rounding.

b. The ethnic groups listed in the table corresponded to those indicated on the patients' identification documents for the surgical procedure. Among the represented ethnic minorities were Mongolians, Manchus, and Huis.

”

Line 493 in Manuscript:

“

patients were required to have undergone central lymph node dissection with a total of at least 5 lymph nodes removed, and (e) there must be a pathologic assessment of FNA for LLNM.

”

3. Decision curve analysis in 6 c,d is based on the assumption that all LLNDs based on true AI predictions are indicated and beneficial to the patient. As outlined in #1, this is probably not

true for patients with microscopic metastases. Due to these considerations, I do not think that the workflow in 7b would be the best clinical management. It is OK to use AI output to have a lower threshold for FNA of the indeterminate lateral neck LNs.

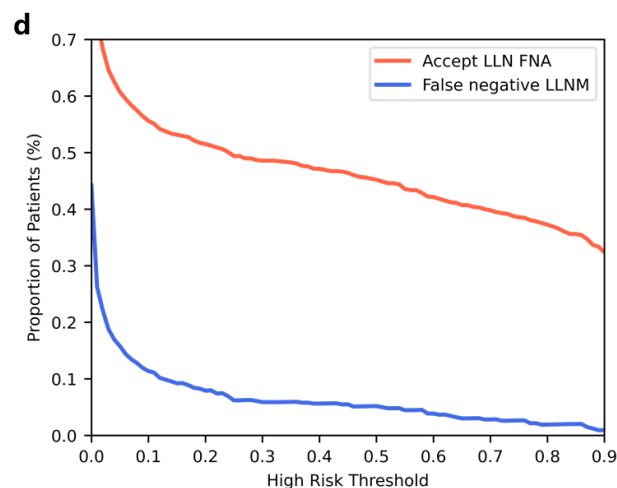
Response:

Your concern is valid, as decision-making should indeed be based on the specific clinical circumstances for patients with minimal or low-risk lymph node metastasis. In response, we have revised the future workflow in Fig. 7b. LLNM-Net will provide a risk probability score and an explanation of high-risk factors for lymph node metastasis based on ultrasound images of thyroid nodules and clinical data. Ultimately, clinical experts will make decisions regarding subsequent diagnostic and treatment methods, integrating their experience with lymph node ultrasound findings, choosing either conservative management or further needle biopsy (FNA). Additionally, we have updated the decision curve presented in Fig. 6d for FNA.

Line 302 in Manuscript:

Fig 6.d

“

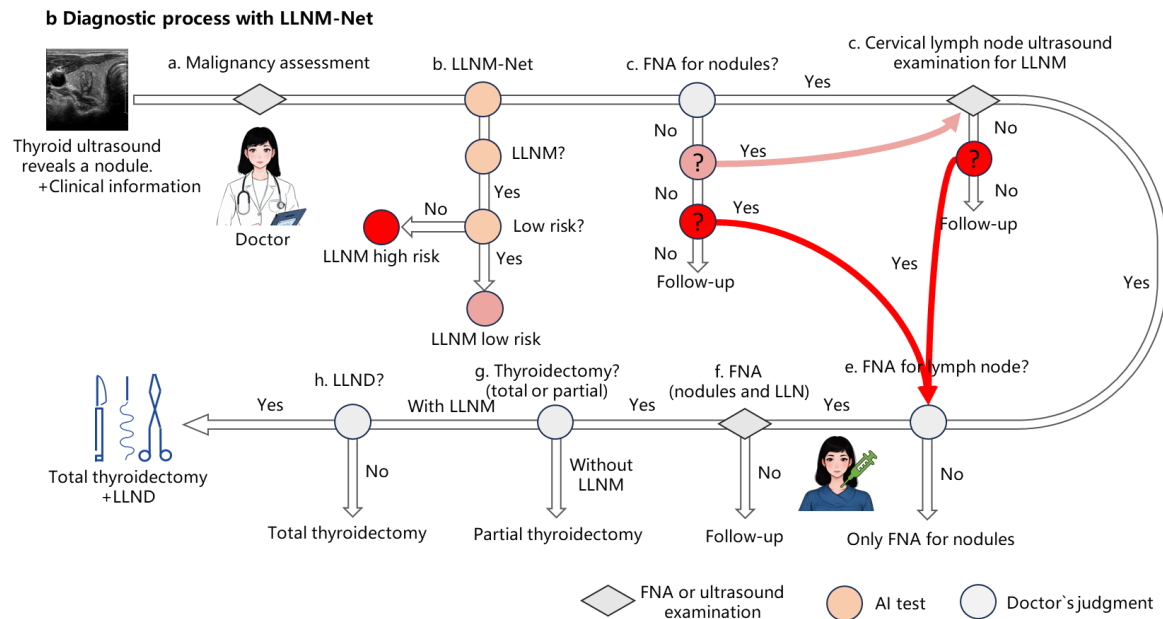


”

Line 353 in Manuscript:

Figure 7.b

“



4. Lines 343-348. LLNM-Net is trained on cancerous nodules only. Most of the thyroid nodules in the population will be benign. Therefore, LLNM-Net cannot be utilized for population screening. Also, detecting lymph node metastases ≥ 3 cm in size is trivial and radiologists would likely achieve close to 100 % sensitivity on such a task. Finally, what is the negative class in Table 2: Both no metastatic disease and < 3 cm metastatic disease?

Response:

Thank you for your insightful comment. We have discussed the malignant thyroid diagnostic model in several previous articles¹²⁻¹⁴. However, as this study focuses primarily on the evaluation and mechanism analysis of lymph node metastasis, we have chosen to emphasize this model in this article. In future practical applications, we can integrate both models to achieve a comprehensive diagnostic workflow, from ultrasound imaging to differentiating benign and malignant status, as well as assessing lymph node metastasis. In response to your suggestion, we have revised the subsequent procedure to recommend FNA biopsy for patients in the medium and high risk stages of metastasis, as shown in Fig. 7.

Regarding the categorization of lymph node metastasis, based on the literature and guidelines¹⁵⁻¹⁸, we have classified the stages as follows:

- Stage 1: Low risk. Five or fewer micro-metastases (<0.2 cm in largest dimension).
- Stage 2: Medium risk. More than five metastatic lymph nodes, and any metastatic lymph node <3 cm in largest dimension.
- Stage 3: High risk. Any metastatic lymph node >3 cm in largest dimension.

We conducted an evaluation experiment to calculate the average maximum diameter of missed positive lymph nodes. The results, shown in Table S3, indicate that for patients in the high-risk stage, a small number of lymph nodes were still missed by doctors. We have made corresponding revisions to Table 2 to show the average classification accuracy for the high, medium, and low stages. Table 2 originally presented binary classification results, with the

negative class being patients in the medium and low-risk stages of metastasis (i.e., metastatic disease with lymph nodes smaller than 3 cm).

Line 340 in Manuscript:

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The model is capable of predicting the stage of lateral lymph node metastasis. Based on clinical guidelines¹²⁻¹⁵, we classified lymph node metastasis into three stages according to the number and size of metastatic nodes:

- Stage 1: Low risk. Five or fewer micro-metastases (<0.2 cm in largest dimension).
- Stage 2: Medium risk. More than five metastatic lymph nodes, and any metastatic lymph node <3 cm in largest dimension.
- Stage 3: High risk. Any metastatic lymph node >3 cm in largest dimension.

Table 2 demonstrates that the model exhibits outstanding predictive performance, achieving an average AUC of 0.971 in external test sets. This suggests that LLNM-Net can precisely identify individuals at medium to high risk, offering physicians reference advice for FNA testing, and ensure regular follow-ups for low-risk individuals, thereby optimizing the efficient allocation of medical resources.

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Line 890 in Manuscript:

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Table 2 Performance of multi-level risk of lymph node metastasis^a.

Cohorts (Cut off = 0.5)	Training set (N=13166)	Validation set (N=3458)	External test sets (N=4060)
AUC	0.984 (0.982-0.987)	0.994 (0.992-0.996)	0.971 (0.964-0.979)
Specificity	0.962 (0.958-0.967)	0.967 (0.960-0.974)	0.938 (0.930-0.948)
Sensitivity	0.908 (0.899-0.918)	0.903 (0.887-0.920)	0.863 (0.845-0.884)
Accuracy	0.938 (0.932-0.944)	0.945 (0.936-0.955)	0.910 (0.896-0.923)
NPV	0.961 (0.956-965)	0.976 (0.971-0.981)	0.953 (0.946-0.961)
PPV	0.904 (0.895-0.914)	0.914 (0.897-0.932)	0.894 (0.878-0.912)

a. The table displays the average values of multi-classification metrics.

”

5. Fig 7a. The best clinical practice is to evaluate central and lateral neck lymph nodes before the FNA (and biopsy suspicious LNs at the time of the procedure, see Recommendation 6 in the ATA guidelines). Also, lateral neck evaluation may change decisions on whether to proceed with thyroidectomy (i.e., the presence of metastatic LNs in the lateral neck would require total thyroidectomy).

Response:

Thank you very much for your suggestion. We have updated the procedure in Fig. 7a. Before FNA, the physician will evaluate the likelihood of lateral lymph node metastasis based on ultrasound detection and perform a biopsy of suspicious lymph nodes during FNA for nodules. Additionally, we have included an assessment of how lateral cervical lymph node evaluation can impact decisions regarding the type of thyroidectomy.

Our AI model is designed to assist physicians in making judgments, reduce false negatives, and improve diagnostic accuracy. It aids in determining the risk of lateral region metastasis, improving the success rate of lateral region puncture diagnostics; it can reduce the rate of missed diagnosis of positive lateral lymph nodes before surgery, and minimizes postoperative recurrence and reoperation in thyroid cancer patients. However, this model will not alter the surgical approach. The AI model’s primary goal is to enhance the diagnostic accuracy of lateral lymph nodes, providing standardized and precise auxiliary testing methods. In Fig. 7b, we demonstrate the improvements in the procedure after incorporating the AI model.

Line 352 in Manuscript:

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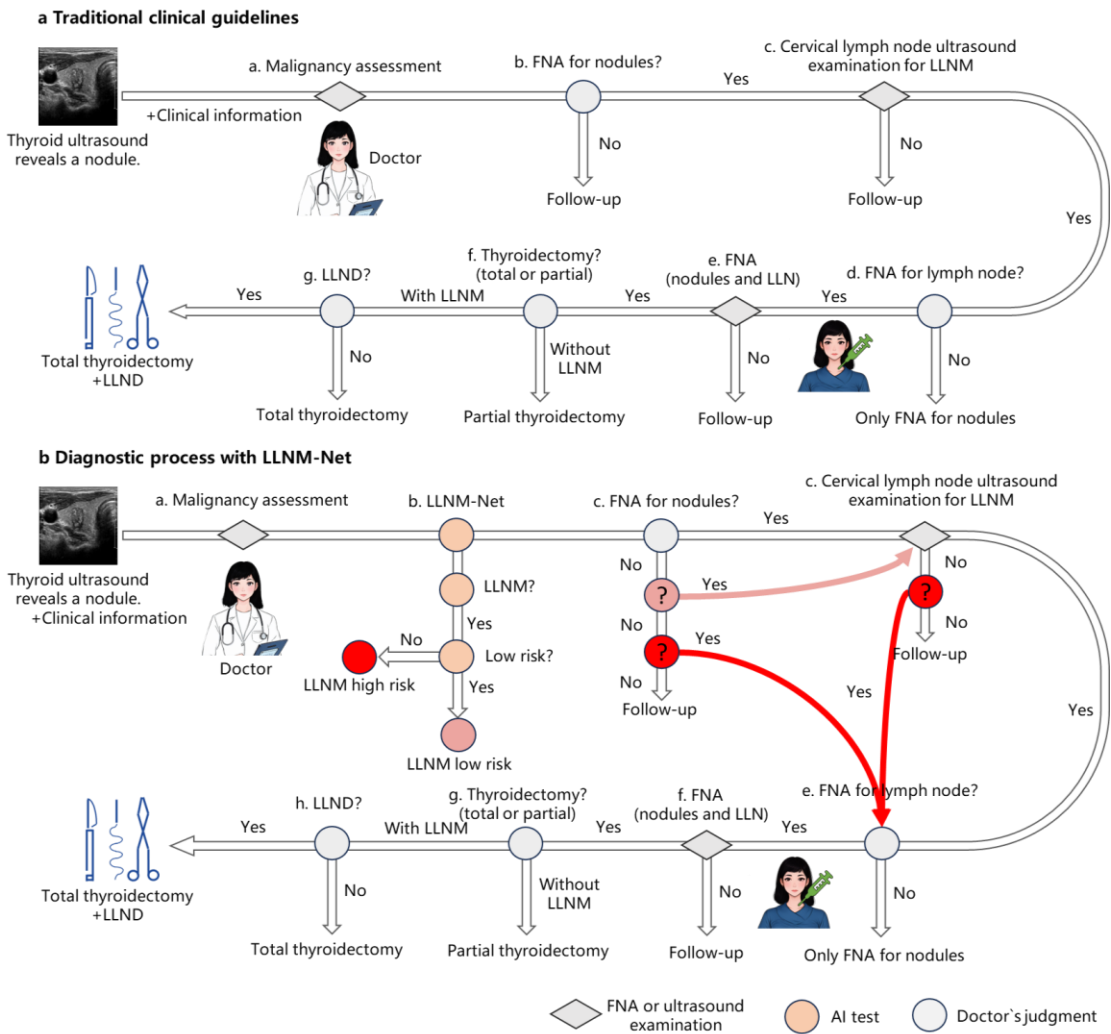


Fig 7. Clinical application of LLNM-Net.

- a.** Traditional clinical guidelines for LLNM. The process primarily involves the following steps: Doctors assess the malignancy of thyroid nodules via ultrasound and clinical information -- Determine whether to perform FNA for nodules -- Cervical lymph node ultrasound examinations for LLNM -- Determine whether to perform FNA for LLN -- FNA for lymph nodes and LLN, among other procedures to assess the necessity of thyroidectomy and LLND.
- b.** Clinical guidelines for LLNM with LLNM-Net. LLNM-Net assists in preoperative risk prediction for

LLNM, guiding the detection of cervical lymph nodes in low-risk patients and recommending FNA for lymph nodes for medium and high risk patients.

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Minor considerations.

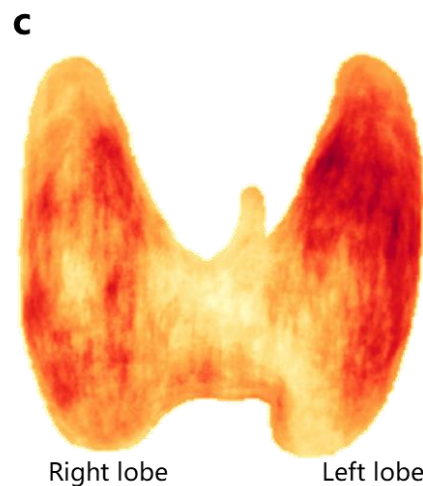
1.Fig 4c. Please label left and right. Is it a radiologist’s view (the right is on the left) or a conventional view?

Response:

Thank you for pointing this out. We have added the labels "Right lobe" and "Left lobe" in Fig. 4c. This image represents the radiologist’s view, or in other words, the three-dimensional thyroid model observed from the front of the patient.

Line 249 in Manuscript:

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